8.5 Practice: estimating and the calculator

← Unit overview · Answers

These sets start easy and build up. In this unit “about right” often **is** right. There is no clock. You need a calculator only for Sets D and E.

Set A: Rounding (*afronden*) (Warm-up)

Round each number. Picture the number line if it helps: which end is it nearer to?

1. 47 to the nearest ten
2. 382 to the nearest hundred
3. 1 649 to the nearest thousand
4. 3,7 to a whole number
5. 12,49 to a whole number
6. €4,95 to whole euros
7. 2,346 to one decimal
8. 7,85 to one decimal
9. 15 500 to the nearest thousand
10. 0,666 to two decimals
11. 63 to the nearest ten

Talk about it: Which ones were exactly halfway? What do we do then?

Set B: Estimate (*schatten*) (Building)

Don’t calculate exactly. Round to friendly numbers and give an estimate. Write down the friendly sum you used.

1. 49 + 31
2. 198 + 403
3. 6 × 49
4. 21 × 19
5. 812 : 4
6. €2,95 + €4,10 + €1,98
7. 3 980 : 20
8. 52 × 11
9. You buy 5 things that each cost €2,99. Is €20 enough?
10. 0,49 × 300

Talk about it: For number 9, did you need an exact answer to say yes or no?

Set C: Does it make sense? (*Klopt het?*) (Building)

A calculator (or a friend in a hurry) gave these answers. Some are right and some are wrong. Use an estimate to decide. If it’s wrong, write what it should be about.

1. 38 × 21 = 798
2. 456 + 389 = 7 145
3. 1 200 : 4 = 300
4. 19,95 × 3 = 598,5
5. 2 500 : 50 = 5

6. 72 × 48 = 3 456
7. 9,9 + 0,1 = 10
8. 15% of €80 = €120
9. “The door of our classroom is 2 cm high.”
10. “28 children go on a trip in cars that take 4 children each, so we need 7 cars.”

Talk about it: Which wrong answer was easiest to catch? What gave it away?

Set D: Head, paper or calculator? (Building → Stretch)

For each sum, first write **H** (head), **P** (paper) or **C** (calculator). Then solve it that way. For the calculator ones, write an estimate first.

1. 100 − 1
2. 25 × 4
3. 3 648 : 12
4. 487 + 235
5. 1,5 × 8
6. 17,8% of €2 345 (round to cents)
7. 999 + 999
8. How many hours are in a year of 365 days?
9. 50% of 846
10. €12,49 × 7

Talk about it: Did you and your parent choose the same letter every time? Whose reason was best for number 8?

Set E: Reading the calculator (Stretch → finish on a win)

Use a calculator. Then decide what the answer **means**.

1. 250 children go on a trip in minibuses with 8 seats each. The calculator says 250 : 8 = 31.25. How many minibuses do you need?
2. Four friends share a restaurant bill of €47,60 equally. The calculator shows 11.9. How much does each friend pay? Write it as money.
3. Type 1 : 3. What does the screen show? Round it to two decimals.
4. Three friends share €20. How much is that each, rounded to cents? What’s the problem?
5. Type 3 + 4 × 2. What does your calculator show? What is the correct answer when you multiply first (as in secondary school)?
6. Apples cost €3,98 per kilo. You buy 2,5 kg. What do you pay? (Round to cents.)
7. 365 days : 7 = ? How many **full** weeks are there in a year?
8. 100 eggs are packed in boxes of 8. How many boxes do you need? How many boxes are **completely** full?
9. Type 10 : 2. What does it show?

Talk about it: In numbers 1 and 8, why is the calculator’s answer not the final answer?

Puzzle corner (optional, just for fun)

1. Without a calculator: is 99 × 99 more or less than 10 000? By how much? (Hint: 100 × 99 is easy.)
2. Estimate: how many times does your heart beat in a day? (Count your pulse for 15 seconds first, sitting still.)

3. Type 12345679 × 9 into a calculator. Then try 12345679 × 18 and × 27. What's going on?